Michael Huang
EN.625.633.81
Homework 6

1.

Example 2.7. Example 2.2 did not provide a general algorithm to simulate beta $\mathcal{B}e(\alpha, \beta)$ random variables. We can, however, construct a toy algorithm based on the Accept–Reject method, using as the instrumental distribution the uniform $\mathcal{U}_{[0,1]}$ distribution when both α and β are larger than 1. (The generic `rbeta` function does not impose this restriction.)

The upper bound M is then the maximum of the beta density, obtained for instance by `optimize` (or its alias `optimise`):

```
> optimize(f=function(x){dbeta(x,2.7,6.3)},
+ interval=c(0,1),max=T)$objective
[1] 2.669744
```

Since the candidate density g is equal to one, the proposed value Y is accepted if $M \times U < f(Y)$, that is, if $M \times U$ is under the beta density f at that realization. Note that generating $U \sim \mathcal{U}_{[0,1]}$ and multiplying by M is equivalent to generating $U \sim \mathcal{U}_{[0,M]}$. For $\alpha = 2.7$ and $\beta = 6.3$, an alternative `R` implementation of the Accept–Reject algorithm is

```
> Nsim=2500
> a=2.7;b=6.3
> M=2.67
> u=runif(Nsim,max=M)     #uniform over (0,M)
> y=runif(Nsim)           #generation from g
> x=y[u<dbeta(y,a,b)]     #accepted subsample
```

and the left panel in Figure 2.4 shows the results of generating 2500 pairs (Y, U) from $\mathcal{U}_{[0,1]} \times \mathcal{U}_{[0,M]}$. The black dots $(Y, Ug(Y))$ that fall under the density f are those for which we accept $X = Y$, and we reject the grey dots $(Y, Ug(Y))$ that fall outside. It is again clear from this graphical representation that the black dots are uniformly distributed over the area under the density f. Since the probability of acceptance of a given simulation is $1/M$ (Exercise 2.5), with $M = 2.67$ we accept approximately $1/2.67 = 37\%$ of the values. ◀

Exercise 6.7 Referring to Example 2.7, consider a $\mathcal{B}e(2.7, 6.3)$ target density.

a. Generate Metropolis–Hastings samples from this density using a range of independent beta candidates from a $\mathcal{B}e(1, 1)$ to a beta distribution with small variance. (*Note:* Recall that the variance is $ab/(a+b)^2(a+b+1)$.) Compare the acceptance rates of the algorithms.
b. Suppose that we want to generate a *truncated* beta $\mathcal{B}e(2.7, 6.3)$ restricted to the interval (c, d) with $c, d \in (0, 1)$. Compare the performance of a Metropolis–Hastings algorithm based on a $\mathcal{B}e(2, 6)$ proposal with one based on a $\mathcal{U}(c, d)$ proposal. Take $c = .1, .25$ and $d = .9, .75$.

Exercise 6.9 Calculate the mean of a gamma $\mathcal{G}(4.3, 6.2)$ random variable using

a. Accept–Reject with a gamma $\mathcal{G}(4, 7)$ candidate;
b. Metropolis–Hastings with a gamma $\mathcal{G}(4, 7)$ candidate;
c. Metropolis–Hastings with a gamma $\mathcal{G}(5, 6)$ candidate.

In each case, monitor the convergence across iterations.

(a)

(b)

2.

We need to calculate the mean of a gamma $G(4.3, 6.2)$ random variable. We know the true mean by definition to be $\frac{n}{\lambda} = \frac{4.3}{6.2} = 0.69354838709$.

(a)

For the accept-reject method with a gamma $G(4, 7)$ candidate, we need to first calculate constant M such that $M \geq sup_x[\frac{f(x)}{g(x)}]$:

$$\frac{f(x)}{g(x)} = \frac{G(4.3, 6.2)}{G(4, 7)}$$

which we can use code to try and investigate, and then proceed:

```
> x_seq <- seq(0.01, 20, length.out = 100)
>
> # Calculate ratio of f(x) / g(x) at each point
```

```
> ratios <- dgamma(x_seq, shape=4.3, rate=6.2) / dgamma(x_seq, shape=4,
rate=7)
> print(ratios)
  [1] 1.825208e-01 5.361883e-01 7.703168e-01 1.019992e+00 1.305276e+00
1.639118e+00
  [7] 2.033787e+00 2.502597e+00 3.060751e+00 3.725963e+00 4.519063e+00
5.464642e+00
 [13] 6.591792e+00 7.934962e+00 9.534960e+00 1.144013e+01 1.370777e+01
1.640574e+01
 [19] 1.961443e+01 2.342908e+01 2.796244e+01 3.334805e+01 3.974399e+01
4.733737e+01
 [25] 5.634968e+01 6.704297e+01 7.972734e+01 9.476962e+01 1.126037e+02
1.337429e+02
 [31] 1.587941e+02 1.884748e+02 2.236337e+02 2.652737e+02 3.145806e+02
3.729556e+02
 [37] 4.420550e+02 5.238358e+02 6.206104e+02 7.351110e+02 8.705652e+02
1.030786e+03
 [43] 1.220276e+03 1.444356e+03 1.709307e+03 2.022547e+03 2.392837e+03
2.830519e+03
 [49] 3.347805e+03 3.959112e+03 4.681459e+03 5.534937e+03 6.543258e+03
7.734414e+03
 [55] 9.141436e+03 1.080331e+04 1.276604e+04 1.508392e+04 1.782100e+04
2.105287e+04
 [61] 2.486871e+04 2.937372e+04 3.469203e+04 4.097006e+04 4.838055e+04
5.712723e+04
 [67] 6.745043e+04 7.963362e+04 9.401111e+04 1.109772e+05 1.309969e+05
1.546186e+05
 [73] 1.824890e+05 2.153706e+05 2.541627e+05 2.999255e+05 3.539092e+05
4.175878e+05
 [79] 4.926992e+05 5.812922e+05 6.857822e+05 8.090170e+05 9.543534e+05
```

```
1.125749e+06
 [85] 1.327868e+06 1.566209e+06 1.847254e+06 2.178642e+06 2.569378e+06
3.030074e+06
 [91] 3.573240e+06 4.213616e+06 4.968577e+06 5.858598e+06 6.907810e+06
8.144647e+06
 [97] 9.602621e+06 1.132122e+07 1.334697e+07 1.573471e+07
>
> # The ratios seem to continue to infinity,
> # and took a long time to run when taking this M,
> # since the acceptance rate becomes extremely low.
> # Instead, I used G(4,6) with a heavier tail to better model
>
> ratios <- dgamma(x_seq, shape=4.3, rate=6.2) / dgamma(x_seq, shape=4,
rate=6)
> print(ratios)
  [1] 0.33477770 0.80365403 0.94347152 1.02085489 1.06752266 1.09545000
1.11069626
  [8] 1.11683343 1.11617694 1.11032784 1.10044584 1.08740073 1.07186240
1.05435774
 [15] 1.03530806 1.01505474 0.99387737 0.97200691 0.94963546 0.92692364
0.90400631
 [22] 0.88099703 0.85799163 0.83507098 0.81230340 0.78974647 0.76744862
0.74545045
 [29] 0.72378581 0.70248271 0.68156415 0.66104871 0.64095122 0.62128316
0.60205318
 [36] 0.58326740 0.56492975 0.54704229 0.52960539 0.51261803 0.49607790
0.47998165
 [43] 0.46432497 0.44910278 0.43430930 0.41993816 0.40598253 0.39243515
0.37928841
 [50] 0.36653446 0.35416519 0.34217236 0.33054757 0.31928235 0.30836817
```

```
0.29779649
 [57] 0.28755876 0.27764647 0.26805114 0.25876439 0.24977788 0.24108341
0.23267284
 [64] 0.22453819 0.21667159 0.20906530 0.20171173 0.19460342 0.18773308
0.18109356
 [71] 0.17467787 0.16847917 0.16249077 0.15670618 0.15111901 0.14572307
0.14051232
 [78] 0.13548087 0.13062298 0.12593307 0.12140573 0.11703566 0.11281774
0.10874699
 [85] 0.10481857 0.10102776 0.09737001 0.09384089 0.09043610 0.08715147
0.08398296
 [92] 0.08092666 0.07797875 0.07513557 0.07239354 0.06974922 0.06719926
0.06474041
 [99] 0.06236955 0.06008363
> M <- max(ratios)
> print(M)
[1] 1.116833
>
> accept_reject_gamma = function(n, M) {
+
+   simulated_values = rep(0, n)
+   simulated_successes = 0
+   simulated_trials = 0
+
+   # Target distribution: Gamma(4.3, 6.2)
+   target_shape = 4.3
+   target_rate = 6.2
+
+   # Candidate distribution: Gamma(4, 6) -- not Gamma(4, 7)
+   cand_shape = 4
```

```
+   cand_rate = 6
+
+   while (simulated_successes < n) {
+
+     simulated_trials = simulated_trials + 1
+
+     x_cand = rgamma(1, shape=cand_shape, rate=cand_rate)
+     uniform = runif(1, 0, M * dgamma(x_cand, shape=cand_shape, rate=cand_ra
+
+     # Accept if uniform < f(x_cand) where f is Gamma(4.3, 6.2)
+     if (uniform < dgamma(x_cand, shape=target_shape, rate=target_rate))
{
+       simulated_successes = simulated_successes + 1
+       simulated_values[simulated_successes] = x_cand
+     }
+
+     if (simulated_trials %% 10 == 0) {
+       print(paste("mean after", simulated_trials, "simulated_trials:",
mean(simulated_values)))
+     }
+   }
+
+   accept_prob = simulated_successes / simulated_trials
+   print(paste("Acceptance probability:", accept_prob))
+   print(paste("Estimated mean:", mean(simulated_values)))
+   hist(simulated_values, density = 50, breaks = 50, main = "2. (a): Accept-
+ }
>
> accept_reject_gamma(100, M)
[1] "mean after 10 simulated_trials: 0.057361156802528"
```

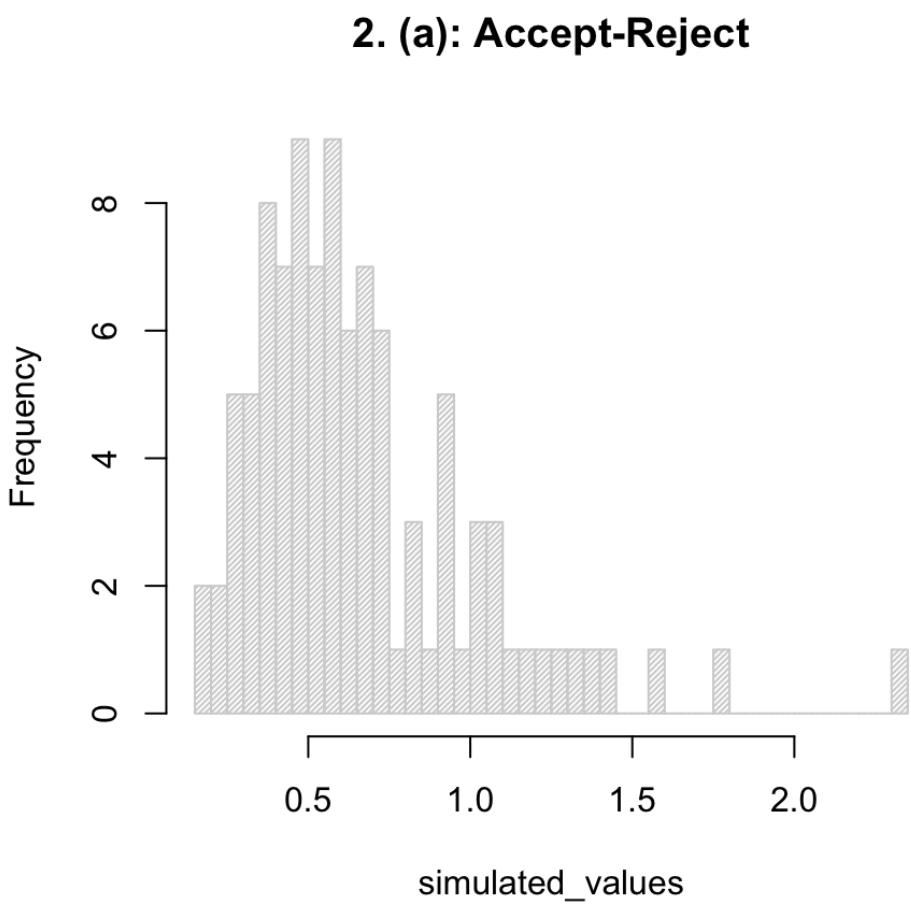
```
[1] "mean after 20 simulated_trials: 0.124639035681143"
[1] "mean after 30 simulated_trials: 0.174761781387247"
[1] "mean after 40 simulated_trials: 0.238516844987886"
[1] "mean after 50 simulated_trials: 0.301053141532097"
[1] "mean after 60 simulated_trials: 0.369637321321696"
[1] "mean after 70 simulated_trials: 0.446326576448048"
[1] "mean after 80 simulated_trials: 0.497404487331674"
[1] "mean after 90 simulated_trials: 0.534486438923785"
[1] "mean after 100 simulated_trials: 0.592475025527754"
[1] "mean after 110 simulated_trials: 0.653229654547747"
[1] "Acceptance probability: 0.900900900900901"
[1] "Estimated mean: 0.664852717498268"
```

With the new candidate gamma $G(4, 6)$, it converges very well. We can see the above means getting better over time.

(b)

Metropolis-Hastings with a gamma $G(4, 7)$ candidate:

```
> met_hastings_gamma = function(n, prop_shape, prop_rate) {
+
+   x = rep(0, n)
+   j = 1
+   acc = 0
+   x[1] = 2
+
+   while (j < n) {
+
+     # Sample candidate value
+     x_cand = rgamma(1, shape = prop_shape, rate = prop_rate)
+
+     # Calculate acceptance probability
```

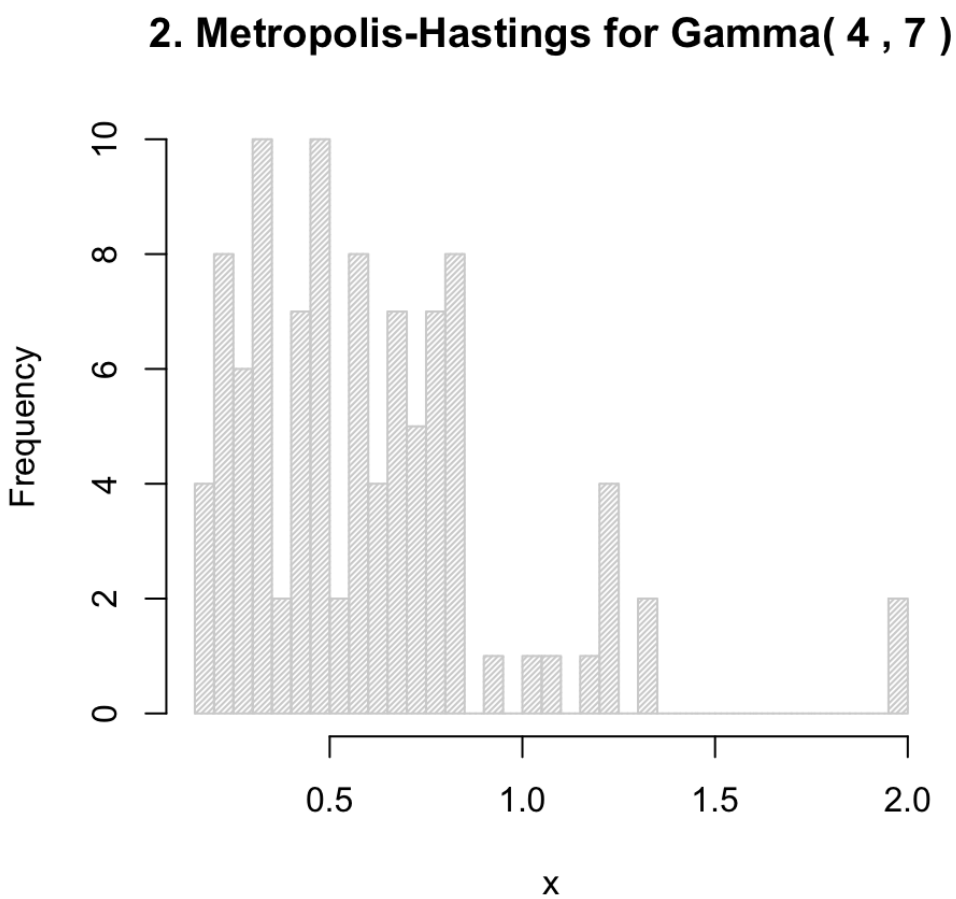



```
+     acc_prob = acc_prob_calc_gamma(x_cand, x[j], prop_shape, prop_rate)
+
+     # Decide whether to accept or not
+     acc_or_not = rbinom(1, 1, acc_prob);
+
+     # if accept or not
+     if (acc_or_not == 1) {
+       acc = acc + 1;
+       j = j + 1;
+       x[j] = x_cand
+     }
+     else {
+       j = j + 1
```

```
+       x[j] = x[j-1]
+     }
+
+     if (j %% 10 == 0) {
+       print(paste("mean after", j, "iterations:", mean(x[1:j])))
+     }
+   }
+
+   acc_probability = acc/n;
+   print(paste("Acceptance probability:", acc_probability));
+   print(paste("Estimated mean:", mean(mean(x[1:n]))))
+   hist(x, breaks = 50, density = 50, main = paste(
+     "2. Metropolis-Hastings for Gamma(", prop_shape, ",", prop_rate,
")")
+     );
+ }
>
> acc_prob_calc_gamma = function(candidate, former, prop_shape, prop_rate)
{
+
+   if (candidate < 0) {
+     acc_probability = 0;
+   }
+   else {
+     # Target densities
+     dens_candidate = dgamma(candidate, shape = 4.3, rate = 6.2)
+     dens_former = dgamma(former, shape = 4.3, rate = 6.2)
+
+     # Proposal densities
+     g_former_given_candidate = dgamma(former, shape = prop_shape, rate
```

```
= prop_rate)
+     g_candidate_given_former = dgamma(candidate, shape = prop_shape,
rate = prop_rate)
+
+     acc_probability = (dens_candidate * g_former_given_candidate) / (dens_
* g_candidate_given_former);
+   }
+
+   acc_probability = min(1, acc_probability);
+   return(acc_probability);
+ }
>
> # 2. (b): Using Gamma(4, 7)
> met_hastings_gamma(100, 4, 7)
[1] "mean after 10 iterations: 0.784790016282701"
[1] "mean after 20 iterations: 0.664775095539249"
[1] "mean after 30 iterations: 0.65066042996016"
[1] "mean after 40 iterations: 0.617259564762832"
[1] "mean after 50 iterations: 0.659536698996242"
[1] "mean after 60 iterations: 0.633522620303361"
[1] "mean after 70 iterations: 0.588662600598103"
[1] "mean after 80 iterations: 0.573466093970765"
[1] "mean after 90 iterations: 0.574118140801531"
[1] "mean after 100 iterations: 0.60192485306117"
[1] "Acceptance probability: 0.79"
[1] "Estimated mean: 0.60192485306117"
```

Using $G(4, 7)$, it converges well, but not as good. We can see the above means getting better over time, although it slightly underestimates.



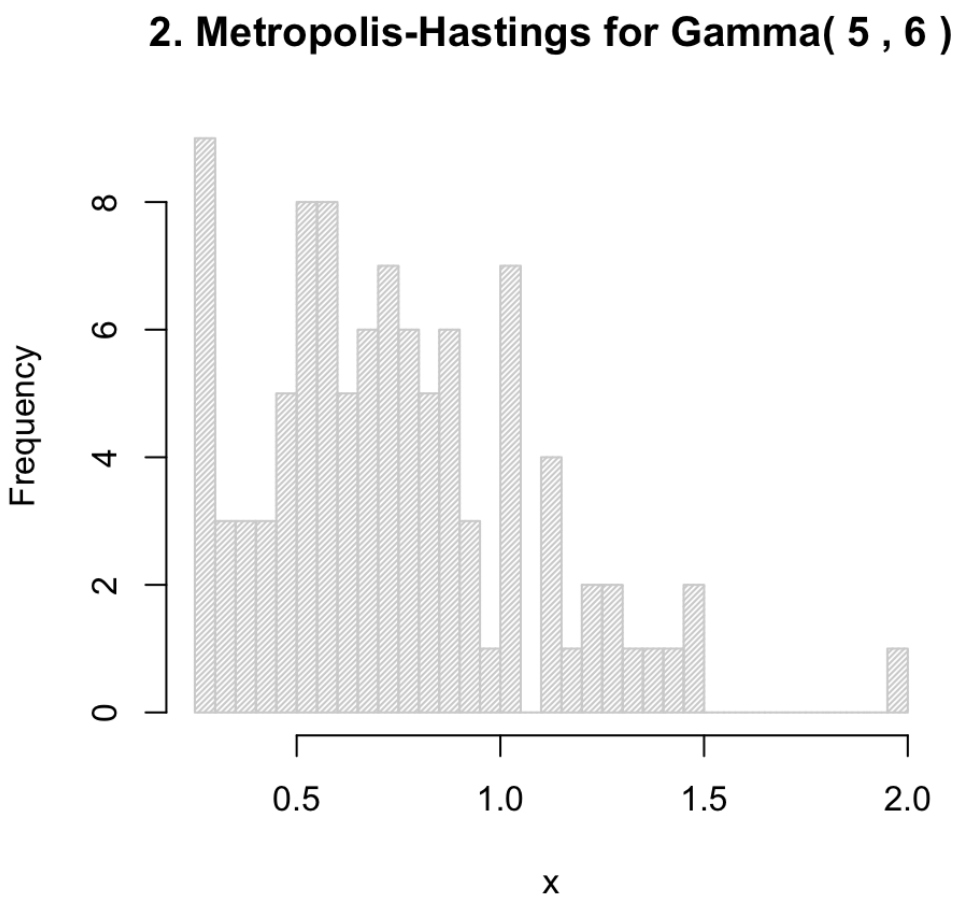
(c)

Metropolis-Hastings with a gamma $G(5, 6)$ candidate:

```
> # 2. (c): Using Gamma(5, 6)
> met_hastings_gamma(100, 5, 6)
[1] "mean after 10 iterations: 0.951377290724033"
[1] "mean after 20 iterations: 0.730853024169404"
[1] "mean after 30 iterations: 0.730709995413933"
[1] "mean after 40 iterations: 0.768977295580612"
[1] "mean after 50 iterations: 0.784397849443638"
[1] "mean after 60 iterations: 0.78316770471714"
[1] "mean after 70 iterations: 0.733693425666811"
[1] "mean after 80 iterations: 0.745312775744982"
```

```
[1] "mean after 90 iterations: 0.743769769713237"
[1] "mean after 100 iterations: 0.742317336251595"
[1] "Acceptance probability: 0.76"
[1] "Estimated mean: 0.742317336251595"
```

Using $G(4,7)$, it also converges very well. We can see the above means getting better over time, and it is slightly closer than using $G(4,7)$, although it slightly overestimates now.



3.

We show the calculations leading to the kernel expression:

$$K(X_t, X_{t+1}) = \rho(X_t, X_{t+1})\, g(X_{t+1}|X_t) + \delta(X_{t+1} - X_t)\,(1 - r(X_t)).$$

We know that

$$K(X_t, X_{t+1}) = K(X_t|X_{t+1})$$

To be thorough (and restating the reasoning behind what's given in the notes), we know by definition that the chain either moves to the candidate or doesn't change; i.e. our next state X_{t+1} is either:

- X_{cand} with probability $\rho(X_t, X_{cand})$
- X_t with probability $1 - \rho(X_t, X_{cand})$

which translates to the conditional probability density being

$$f(X_{t+1}|X_{cand}, X_t) = \delta(X_{cand} - X_{t+1})\rho(X_t, X_{cand}) + \delta(X_t - X_{t+1})(1 - \rho(X_t, X_{cand}))$$

as given. We then marginalize over all possible candidate values, i.e. taking

$$K(X_t, X_{t+1}) = \int_{\mathbb{R}} f(X_{t+1}|X_{cand} = x_{cand}, X_t) g(X_{cand} = x_{cand}|X_t) dx_{cand}$$

Plugging in:

$$= \int_{\mathbb{R}} [\delta(X_{cand} - X_{t+1})\rho(X_t, X_{cand}) + \delta(X_t - X_{t+1})(1 - \rho(X_t, X_{cand}))] g(x_{cand}|X_t) dx_{cand}$$

(simplifying a bit, for sanity)

$$= \int_{\mathbb{R}} [\delta(x - X_{t+1})\rho(X_t, x) + \delta(X_t - X_{t+1})(1 - \rho(X_t, x))] g(x|X_t) dx$$

$$= \int_{\mathbb{R}} \delta(x - X_{t+1})\rho(X_t, x) g(x|X_t) dx + \int_{\mathbb{R}} \delta(X_t - X_{t+1})(1 - \rho(X_t, x)) g(x|X_t) dx$$

Looking at the first integral, this essentially represents the sum of all the scenarios where we ended up at X_{t+1} – this is due to the product of the delta function representing that we reached the transition or not from the candidate value, the rho function representing the probability of accepting the candidate value, and the g function representing the probability of generating the candidate value. We therefore only care about the actual value of $x_{cand} = X_{t+1}$, since is the only place under the integral where the delta function is 1 (and nonzero). We can therefore simplify the first term accordingly based on this logic:

$$= \rho(X_t, X_{t+1}) g(X_{t+1}|X_t) + \int_{\mathbb{R}} \delta(X_t - X_{t+1})(1 - \rho(X_t, x)) g(x|X_t) dx$$

And continuing to integrate:

$$= \rho(X_t, X_{t+1})g(X_{t+1}|X_t) + \delta(X_t - X_{t+1}) \int_{\mathbb{R}} (1 - \rho(X_t, x))g(x|X_t)dx$$

$$= \rho(X_t, X_{t+1})g(X_{t+1}|X_t) + \delta(X_t - X_{t+1})[\int_{\mathbb{R}} g(x|X_t)dx - \int_{\mathbb{R}} \rho(X_t, x)g(x|X_t)dx]$$

$$= \rho(X_t, X_{t+1})g(X_{t+1}|X_t) + \delta(X_t - X_{t+1})[1 - \int_{\mathbb{R}} \rho(X_t, x)g(x|X_t)dx]$$

We now define $r(X_t) = \int_{\mathbb{R}} \rho(X_t, x)g(x|X_t)dx.$, which represents the combined probability of proposing and accepting some candidate value. We therefore now have

$$= \rho(X_t, X_{t+1})g(X_{t+1}|X_t) + \delta(X_t - X_{t+1})(1 - r(X_t))$$

and because the delta function is symmetric, we have

$$= \rho(X_t, X_{t+1})g(X_{t+1}|X_t) + \delta(X_{t+1} - X_t)(1 - r(X_t))$$

exactly as we sought to find.

4.

We are given the Baker suggested acceptance probability $p = \frac{f_X(x_{i+1})}{f_X(x_{i+1})+f_X(x_i)}$. We aim to find whether $g(x_{i+1}|x_i) = g(x_i|x_{i+1})$ guarantees detailed balance, which is defined in the notes to be $P(X = x_i)P(X_i \to X_{i+1}) = P(X = x_{i+1})P(X_{i+1} \to X_i)$. Let's evaluate this statement by solving either side. Starting with the left side:

$$P(X = x_i)P(X_i \to X_{i+1}) = f(x_i) \cdot g(x_{i+1}|x_i) \cdot \frac{f(x_{i+1})}{f(x_{i+1}) + f(x_i)}$$

$$P(X = x_i)P(X_i \to X_{i+1}) = \frac{f(x_i)f(x_{i+1})g(x_{i+1}|x_i)}{f(x_{i+1}) + f(x_i)}$$

Let's evaluate the right hand side and see where it goes:

$$P(X = x_{i+1})P(X_{i+1} \to X_i) = f(x_{i+1}) \cdot g(x_i|x_{i+1}) \cdot \frac{f(x_i)}{f(x_i) + f(x_{i+1})}$$

Given that $g(x_{i+1}|x_i) = g(x_i|x_{i+1})$:

$$= f(x_{i+1}) \cdot g(x_{i+1}|x_i) \cdot \frac{f(x_i)}{f(x_i) + f(x_{i+1})}$$

$$= \frac{f(x_i)f(x_{i+1})g(x_{i+1}|x_i)}{f(x_{i+1}) + f(x_i)}$$

This is the same as the left hand side, which indicates that this does indeed guarantee detailed symmetry. Intuitively, the probability works since it indicates the proportion of the total probability belongs to the x_{i+1} state, and the g function symmetry means that the proposed state generation doesn't favor either side. So there is equilibrium in the sense that only the acceptance probability matters, which is balanced out by the probability of being in the initial state, whether that is x_i or x_{i+1}.